SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Programme: **B.E / B.Tech**

Course Code: **CSA5102**

Course Name: **Cryptography and Network Security**

Duration: **1: 30 Hr**

Date :

Max Marks:
50 Marks

48/50

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.
CO2	Analyze public-key cryptosystem strategies using RSA and key exchange for strengthening information security.

Answer all the Questions

1. A security analyst at a small firm is asked to demonstrate classical encryption to new trainees. The plaintext "MEETATNOON" must be encrypted using a Caesar cipher with shift 7. Show the encryption and decryption process step by step, first converting each letter to its numeric value (A=0, B=1, ..., Z=25), applying the formulas C = (P + 7) mod 26 and P = (C − 7) mod 26, and then converting the numbers back to letters. Next, encrypt the same plaintext with a Vigenère cipher using the repeating key "SUN", showing the key shift for every letter position. Compare the Caesar cipher with the Vigenère cipher in terms of

(a) key space

(b) treatment of repeated patterns such as the double letters EE and OO and the repeated letters T and N, and

2. A cryptography instructor wants to show how matrix algebra is used in classical block ciphers. Encrypt the plaintext "MATH" using a 2 × 2 Hill cipher with key K = [[3, 2], [5, 7]], using A=0, B=1,..., Z=25. Show the conversion of the plaintext into column vectors (MA and TH), and show the matrix multiplication and modulo-26 reduction for each block to obtain the ciphertext. Determine whether the key matrix is invertible modulo 26 by computing its determinant and checking gcd(det, 26) = 1. Find the modular inverse of the determinant, compute the inverse key matrix K^{-1} using the adjugate method, and verify that $K \cdot K^{-1} \equiv I \pmod{26}$. Explain how K^{-1} enables decryption by applying it to the ciphertext blocks and recovering "MATH". Show why the alternative key K = [[2, 3], [4, 5]] is unacceptable by calculating its determinant and its gcd with 26. Finally, identify and explain a major weakness of the Hill cipher, such as its linearity and vulnerability to known-plaintext attacks.
(10) (CO1) (BL4)

3.a) A hospital network of 200 clinics must send large MRI archives (about 1 GB each) to a central radiology server, and every clinic must also exchange encryption keys with the server securely. The IT team is choosing between a symmetric cipher (AES-256-GCM) and the RSA public-key algorithm. Compare the two in terms of
a.key usage (who holds which key),
b.performance and maximum data size per operation,
c.scalability, including the number of keys needed if all 200 clinics must communicate pairwise versus using RSA key pairs, and key-distribution requirements, including the need for certificates to authenticate public keys. (5) (CO1) (BL4)

b) Using this comparison, design a hybrid encryption workflow that uses each algorithm where it is most appropriate, listing the steps carried out by the sending clinic (certificate validation, DEK and nonce generation, AES-GCM encryption with associated data, RSA-OAEP key wrapping, and packaging) and by the receiving server (key unwrapping, decryption, and tag verification). Also explain how secure key storage should be handled (HSM/KMS, non-exportable keys, role-based access, MFA, audit logging, backup and recovery), and how key rotation should be handled (per-file DEKs, periodic rotation of the key-encryption key, re-wrapping, retirement of old keys, and emergency rotation after suspected compromise).
(5) (CO2) (BL4)

4. For an RSA demonstration in a cryptography laboratory, let the primes be p = 11 and q = 19, with public exponent e = 7. Calculate the modulus n and Euler's totient $\varphi(n)$, and confirm that $\gcd(e, \varphi(n)) = 1$. Determine the private exponent d such that $e \cdot d \equiv 1 \pmod{\varphi(n)}$ using the extended Euclidean algorithm, showing every division step and the back-substitution. State the public key and the private key. Encrypt the message M = 5 using C = M^e mod n, showing the intermediate modular exponentiation steps. Verify the result by decrypting the ciphertext using M = C^d mod n (you may use the Chinese Remainder Theorem to simplify the calculation) and confirm that the original message is recovered. Finally, explain why real systems require much larger parameters (2048-bit or 3072-bit moduli) and careful padding such as RSA-OAEP for encryption and RSA-PSS for signatures, discussing factoring attacks, determinism, malleability and small-message attacks on textbook RSA. (10) (CO2) (BL4)

5. A financial institution transmits confidential customer information across a network and must protect it against unauthorized access. Apply suitable classical encryption techniques, such as Caesar cipher, Playfair cipher, and Vigenère cipher, to encrypt and decrypt the given plaintext. Extend your solution by applying an appropriate symmetric or asymmetric cipher to secure the same information. Demonstrate the encryption and decryption procedures, compare the techniques in terms of key usage and security, and justify which approach is more suitable for protecting sensitive financial data. (10) (CO1) (BL3)

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Course : CSA 5102 Cryptography and Network Security

1. Symmetric cryptosystem selection and modes of operation

1. caser cipher

Gin :

plaintext : meet at noon

key : 7

Encryption $c = (p+k) \bmod 26$

Decryption $p = (c-k) \bmod 26$

Here :

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

W	X	Y	Z
22	23	24	25

$M \rightarrow (12+7) \bmod 26 = 19 \rightarrow T$

$E \rightarrow (4+7) \bmod 26 = 11 \rightarrow L$

$E \rightarrow (4+7) \bmod 26 = 11 \rightarrow L$

$T \rightarrow (19+7) \bmod 26 = 0 \rightarrow A$

$A \rightarrow (0+7) \bmod 26 = 7 \rightarrow H$

$T \rightarrow (19+7) \bmod 26 = 0 \rightarrow A$

$N \rightarrow (13+7) \bmod 26 = 20 \rightarrow U$

$O \rightarrow (14+7) \bmod 26 = 21 \rightarrow V$

$O \rightarrow (14+7) \bmod 26 = 21 \rightarrow V$

$N \rightarrow (13+7) \bmod 26 = 20 \rightarrow U$

Encrypted text : TLLAHAUVVU

Decryption:

T → (19-7) mod 26 = 12 → M

L → (11-7) mod 26 = 4 → E

L → (11-7) mod 26 = 4 → E

A → (0-7) mod 26 = 19 → T

Decrypted text = MEETATNOON

2. vigenere cipher

Gin : plaintext : MEETATNOON

Key : SUN

The key is repeated until it matches plaintext length

Repeated key : SUNSUNSUNS

plaintext : 12, 4, 4, 19, 0, 19, 13, 14, 14, 13

key : 18, 20, 13, 18, 20, 13, 18, 20, 13, 18

* M+S = (12+18) mod 26 = 4 = E
* E+U = (4+20) mod 26 = 24 = Y
* E+N = (4+13) mod 26 = 17 = R
* T+S = (19+18) mod 26 = 11 = L

Encrypted text: EYRLUGFIBF

* caser cipher uses only one fixed shift value
* vigenere cipher uses multiple key values based on a repeating keyword.
* caser has only 26 possible shifts
* A three letter vignere key $26^3 = 17,576$ possible key.

cipher produces same encrypted letter whenever the same plaintext letter occur

* vigenere can encrypt same plaintext letter differently depending key position.

Caser cipher weakness

* It has very small key space
* It can be broken by trying all possible shifts.
* Letter frequency analysis can identify the original message.

vigenere cipher weakness

* Repeating keys create patterns
* Kasiski examination can help identify the key length
* Frequently analysis can be applied after identifying the key length.

Recommendation

vigenere is better than caesar for an educational demonstration because it was multiple shifts and provide better protection against simple frequency analysis. However. neither cipher is suitable for modern confidential patient record

Modern Encryption AES 256 GCM

AES 256 GCM is modern symmetric encryption algorithm

* It uses 256 bit Secret key
* It is suitable encryption large amount of data

* GCM provide confidentiality and Integrity verifi
* A unique nonce must be used with each encrypt
under the same key.

2. Hill cipher Encryption and decryption:

Gin: plaintext: MATH

Keymatrix $K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

The plaintext is divided into two pair:

* MA
* TH

Encryption process

M=12, A=0

T=19, H=7

$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$

$TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

$C = K \times p \pmod{26}$

$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 36 \\ 60 \end{bmatrix}$

Taking modulo 26:

$= \begin{bmatrix} 10 \\ 8 \end{bmatrix}$

10=K and 8=I

MA becomes KI

Encrypt second pair TH $= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 17 \end{bmatrix} = \begin{bmatrix} 71 \\ 144 \end{bmatrix}$

3. aking modulo 26: $= \begin{bmatrix} 19 \\ 14 \end{bmatrix}$ $19 = T, 14 = O$

TH becomes TO

The determinant of a 2x2 matrix is:

$$|K| = (3 \times 7) - (2 \times 5) = 21 - 10 = 11$$

For the key matrix to be invertible modulo 26:

$$\gcd(11, 26) = 1$$

Since the GCD is 1, key matrix is invertible.

Find the inverse matrix

The inverse of 11 modulo 26 is 19 becomes:

$$11 \times 19 = 209$$

$$209 \bmod 26 = 1$$

The inverse

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

verify the inverse

$$K \times K^{-1} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 27 & 52 \\ 78 & 105 \end{bmatrix}$$

Taking modulo 26: $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The inverse matrix is correct.

Decryption:

$$P = K^{-1} \times C \pmod{26}$$

FOR KI

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 142 \\ 130 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

This give MA

For To $= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} 253 \\ 241 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$

This give TH

check the alternative key

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Determinant:

$$(2 \times 5) - (3 \times 4) = 10 - 12 = -2$$

$$\gcd(-2, 26) = 2$$

Since the GCD is not this matrix has no modular inverse modulo 26.

weakness of hill cipher:

* It uses matrix multiplication for encryption.
* It is linear encrypted method.
* Known plaintext and ciphertext pairs can help attackers discover the key matrix.

* It is not suitable for modern confidential communication

Secure Encryption Strategy Hospital MRI archives:

Part A - comparison of AES-256-Gcm and RSA

A hospital network has 200 clinics that send large MRI archieves of approximately 1GB.

Two encryption algorithm are considered

AES-256-Gcm and RSA

AES 256 Gcm

* It is Symmetric encryption algorithm
* The same secret key used encryption and decryption
* It is very fast and suitable large files
* It is uses 256 bits key.
* Gcm provide data integrity through authentication tags.

RSA

* It is assymetric encryption algorithm
* It uses public key and private key
* The public key encryption or wraps small secret key, wraps it.
* It is slower than AES
* It is not suitable for directly encrypting a 1GB MRI file.

Key management and scalability:

If every pair of 200 clinics shares sepearte AES key the number of keys required is $\frac{200 \times 199}{2} = 19,900$

This makes key management difficult.

* AES 256 Gcm best large MRI files while RSA is useful for securely protecting AES key.

part B: Hybrid Encryption workflow

A hybrid encryption system combines AES and RSA to provide security and perfomance.

* certificate validation
* Generate a secrete key
* Generate a nonce
* Encrypt the MRI file
* protect the AES key
* Send the package
* Decryption

part-C Key management and security

Key protection:

* Store importank key in a hardware security module or key management servic.
* prevent unauthorized users from accessing private key.
* use role based accss control.

Key Rotation:

* Generate a sepearte AES key for each file
* Rotate key encryption keys periodically.
* Re-encrypt or rewarp protected keys when required
* maintain secure backups for recovery.

Encryption and Decryption

$p = 11, q = 19, e = 7$

$m = 5$

$n = p \times q$

$n = 11 \times 19 = 209$

Euler's Totient function

$$\phi(n) = (p-1)(q-1)$$
$$= (11-1)(19-1)$$
$$= 10 \times 18$$
$$= 180$$

check the public Exponent

Given: $e = 7$

For RSA the GCD of e and $\phi(n)$ must be 1.

$$\gcd(7, 180) = 1$$

The public exponent is valid

$$ed \equiv 1 \pmod{180}$$
$$7d \equiv 1 \pmod{180}$$
$$180 = 7 \times 25 + 5$$
$$7 = 5 \times 1 + 2$$
$$5 = 2 \times 2 + 1$$
$$1 = 5 - 2 \times 2$$
$$= 5 - 2(7-5)$$
$$= 3(180 - 25 \times 7) - 2 \times 7$$
$$= 3(180) - 77(7)$$
$$d = -77 \bmod 180$$
$$d = 103$$

public key and private key

Public Key:

$(e, n) = (7, 209)$

Private key:

$(d, n) = (103, 209)$

Encryption:

$c = m^e \bmod n$

$c = 5^7 \bmod 209$

$5^2 = 25$

$5^4 = 625 \bmod 209 = 207$

$c = 5^4 \times 5^2 \times 5 = 207 \times 25 \times 5 \bmod 29$

$c = 168$

Decrypt:

$m = c^d \bmod n$

$168 \bmod 11 = 3$

$d \bmod 10 = 3$

$3^3 \bmod 11 = 5$

$168 \bmod 19 = 16$

$d \bmod 18 = 13$

$16^{13} \bmod 19 = 5$

$m \equiv 5 \pmod{11}$

$m = 5 \pmod{19}$

Encryption techniques for a financial institution

Caesar cipher

Bin

plaintext: BANK DATA

key: 7

$C = (p+K) \bmod 26$

B → (1+7) = 8 = I

A → (0+7) = 7 = H

N → (13+7) = 20 = U

K → (10+7) = 17 = R

D → (3+7) = 10 = K

A → (0+7) = 7 = H

T → (19+7) = 0 = A

A → (0+7) = 7 = H

ciphertext: IHURKAH

playfair cipher

plaintext: BANKDATA

keyword: MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

BA NK DA TA

BA → IB

NK → RG

DA → BR

TA → SR

ciphertext : IBRGBRSR

Vigenere cipher :

plaintext : BANKDATA

key : SUN

Repeated key : SUNSUNSU

$$c = (p+K) \bmod 26$$

B + S = 19 = T

A + U = 20 = O

N + N = 0 = A

K + S = 2 = C

D + U = 23 = X

A + N = 13 = N

T + S = 11 = L

A + U = 20 = U

ciphertext : TUACKNLU

Decrypted : BANKDATA